

LPN / Hamilton–Jacobi Numerics: Protocol, Findings, and Open Items

Internal working note (not for publication)

Audit 2026-07-08; rewritten 2026-07-09

Abstract

This note states the current numerical protocol, the bugs found and fixed, the claims we have retracted, and what remains open. It replaces the 2026-07-08 audit, whose structure had become a body arguing with its own addendum. The chronological record, including every superseded number, lives in `numerics/changes.txt`; nothing here repeats it.

Two summary judgements. First, the reported pipeline is now correct but its *numbers* have moved by orders of magnitude several times, always because the experimental setup was wrong rather than the code: a corrupted target, a training box that starved both networks, a training budget that left every network under-trained, and an architecture that silently departed from the manuscript. Second, on a symmetric protocol the two recovery routes are comparable; the defensible claim is that Route 2 matches a fully-tuned inversion baseline while never inverting, *not* that it dominates.

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1 The protocol, as it now stands

Let J be the prior, $H(\mathbf{p}) = \frac{1}{2} \|\mathbf{p}\|_2^2$, and let $S(\cdot, t)$ solve the Hamilton–Jacobi equation with initial datum J . The first network regresses the convex potential

$$\psi(\mathbf{y}) = \frac{1}{2} \|\mathbf{y}\|_2^2 - S(\mathbf{y}, 1) = \left(J + \frac{1}{2} \|\cdot\|_2^2\right)^*(\mathbf{y}), \quad \text{so} \quad \nabla \psi = \text{prox}_J.$$

Boxes. Errors are reported on the *query box* $[-4, 4]^n$. The potential ψ is trained on a *larger* box $[-A, A]^n$ with $A = \text{problem.train_halfwidth}(4)$, the least A for which $\nabla \psi([-A, A]^n) \supseteq [-4, 4]^n$. Both routes need this. Route 1 evaluates ψ_θ at the preimage $\mathbf{y}^* = (\nabla \psi)^{-1}(\mathbf{x})$, which *leaves* the query box; Route 2 trains its second network G at the inputs $\mathbf{y}_k = \nabla \psi(\mathbf{x}_k)$, whose support lies strictly *inside* it. Per family: $A = a + t$ for **QuadraticL1**; $A = ((1 + \sigma)a + \|\mu\|_\infty)/\sigma$ for **Minplus**; and $A = a$ for **NegL1** ($\mathbf{y}^* = \text{soft}(\mathbf{x}, 1)$) and **ConcaveQuad** ($\mathbf{y}^* = \mathbf{x}/2$), whose preimage maps contract.

Network. Softplus ICNN, $\beta = 5$, two layers, *width 256 at every dimension*, convexity enforced by clipping the inner weights to be non-negative after each step. The default (Kaiming) initialisation is used and is correct: see §2.6.

Training. Adam, mini-batch 512, and a budget of $S = 250\,000$ *optimizer steps* — not epochs. Learning rate 10^{-3} , decayed $\times 0.1$ at 50% and 75% of S (fractions, so the schedule is invariant to S). *No early stopping*; we keep the best-validation checkpoint. Both networks use MSE. $N = 15\,000n$ training points on the training box (seed 1), 4000 validation (seed 2), 4000 test (seed 3).

Recovery, one protocol for both routes. The same 1000 query points; unconditional RMSE for both; no abstention; no box projection for Route 1.

- **Route 1** (baseline): Fang et al.’s regularized inversion, $\min_{\mathbf{y}} \psi(\mathbf{y}) + \frac{\alpha}{2} \|\mathbf{y}\|_2^2 - \langle \mathbf{x}, \mathbf{y} \rangle$, then $J(\mathbf{x}) = \langle \mathbf{x}, \mathbf{y} \rangle - \psi(\mathbf{y}) - \frac{1}{2} \|\mathbf{x}\|_2^2$. Run at each $\alpha \in \{0, 0.1\}$; report the *best* RMSE.
- **Route 2** (reported): one forward pass, $J(\mathbf{x}) = G(\mathbf{x}) - \frac{1}{2} \|\mathbf{x}\|_2^2$, no inner optimization, no α .

Certificates, identical for both routes and free of ground truth. Because $G \approx \psi^*$ we have $\nabla G = (\nabla \psi)^{-1}$, so Route 2 carries an implied preimage $\mathbf{y}_2(\mathbf{x}) = \nabla G(\mathbf{x})$ at no cost. With $\mathbf{y}(\mathbf{x})$ from the solve (Route 1) or from ∇G (Route 2),

$$r(\mathbf{x}) = \frac{\|\nabla \psi_\theta(\mathbf{y}(\mathbf{x})) - \mathbf{x}\|_2}{\max(1, \|\mathbf{x}\|_2)}, \quad \text{FY}(\mathbf{x}) = \psi_\theta(\mathbf{y}(\mathbf{x})) + G_\theta(\mathbf{x}) - \langle \mathbf{x}, \mathbf{y}(\mathbf{x}) \rangle \geq 0.$$

Neither suffices alone. The residual r measures the α -distortion exactly — at an exact regularized solve $r = \alpha \|\mathbf{y}\|_2 / \max(1, \|\mathbf{x}\|_2)$, verified to a few percent — but it is *blind to divergence*: where ψ_θ is asymptotically affine, $\nabla \psi_\theta(\mathbf{y}) \rightarrow \mathbf{x}$ along a diverging minimizing sequence, so $r \approx 0$ is exactly what divergence looks like. The gap FY detects a bad conjugate pair but is budget-invariant. Only doubling the inversion budget tests well-posedness.

What depends on the dimension. Only $N = 15\,000n$ and the training box A (through the family). Width, depth, β , batch, S , the decay schedule, the loss, the evaluation set, and the α grid are all constant. Epochs are a *derived* quantity, $S \cdot 512/N$; they are not a control and are not reported. Two consequences to state in the manuscript: $|J|$ grows with n (e.g. $\mathbb{E} \|\mathbf{x}\|_1 = 2n$ on $[-4, 4]^n$), so **absolute RMSE is not comparable across dimensions**; and N/P ranges from 0.14 at $n = 2$ to 4.6 at $n = 64$, where $P = 2h^2 + 3nh + 3h + n + 1$ is the parameter count.

Artifacts. Every run writes `logs/ckpt/<run>_psi.pth`, `<run>_G.pth` and `logs/<run>metrics.json`. Figures are therefore regenerated from weights rather than from a retrain, which is both cheap (≈ 3 minutes against ≈ 28) and traceable: a figure can always be tied to the checkpoint that produced it.

Operational guard. Runs above $n = 8$ are opt-in (`bin/run_all.py --allow-high-dim`).

Reference point. `quadratic_l1`, $n = 16$, under the protocol above: ψ validation MSE 7.92×10^{-3} , G validation MSE 2.71×10^{-2} , Route 1 RMSE 0.196 ($\alpha = 0$, no divergence, $\max \|\mathbf{y}\|_\infty = 6.59$), Route 2 RMSE 0.284; relative errors 0.61% and 0.89% against $\mathbb{E} \|\mathbf{x}\|_1 = 32$.

2 Bugs found and fixed

#	When	Bug
1	07-08	Corrupted Moreau target in the quadratic- L^1 family (§2.1). Corrupts ψ , hence <i>both</i> routes.
2	07-08	<code>NegL1</code> value function missed the dimension factor: $-t/2$ instead of $-nt/2$. Error exactly $(n-1)t/2$.
3	07-08	Route-1 recovery used $\ \mathbf{x}\ _2$ where the Fenchel identity requires $\frac{1}{2} \ \mathbf{x}\ _2^2$.
4	07-08	The two networks used different losses (ψ : MAE, G : MSE). Both now MSE.
5	07-08	<code>lib/network.py</code> used Mish, breaking the ICNN convexity guarantee. Dropped.
6	07-08	<code>invert_ls</code> is nonconvex and degenerate on the affine pieces of piecewise-linear priors. Retired.
7	07-09	Both max-plus targets carry the wrong sign on $tH(\mathbf{p})$ (§2.2). Retracts the Phase-1 verdict that <code>exp_4_1_3</code> was clean.
8	07-09	The training box equalled the query box, starving both networks near the boundary (§1).
9	07-09	Reported validation MSE described a <i>discarded</i> model: training early-stops and restores the best-validation weights, but the runner reported the last epoch's error.
10	07-09	The 300-epoch budget left every network under-trained (§2.3).
11	07-09	The width tiering 64/128/256 was our own deviation from the manuscript (§2.5). Reverted to a fixed 256.
12	07-09	Early stopping (ours, added with the validation set) truncated the G network on mini-batch noise. Removed; best-validation checkpoint kept (§2.4).
13	07-09	Copying the paper's 20% decay cadence starves the high-learning-rate phase, which is what ψ 's error tracks (§2.4).
14	07-09	Route-1's α selection took a plain minimum, so a <i>diverged</i> solve was reported whenever it happened to score better. Diverged runs are now excluded.

2.1 [fixed] The Moreau target

For $J = \|\cdot\|_1$ and $t = 1$, $S(\mathbf{y}, 1)$ is the separable Huber function. The notebooks returned $\sum_i |\text{prox}(y_i)| + nt/2$ with n hardcoded to 1: they dropped the curvature term $\frac{1}{2}(\text{prox} - \mathbf{y})^2$ and added a dimension-independent constant. Corrected in `src/targets.py`.

2.2 [fixed] The max-plus targets

For $J(\mathbf{x}) = \max_i \{\langle \mathbf{p}_i, \mathbf{x} \rangle - \gamma_i\}$ the two notebooks use $S(\mathbf{y}, 1) = \max_i \{\langle \mathbf{p}_i, \mathbf{y} \rangle + \frac{1}{2} \|\mathbf{p}_i\|_2^2\}$ ($\gamma = 0$) and $S(\mathbf{y}, 1) = \max_i \langle \mathbf{p}_i, \mathbf{y} \rangle$ ($\gamma_i = \frac{1}{2} \|\mathbf{p}_i\|_2^2$). Both *exceed* J at every test point, so neither is a Moreau envelope, hence neither is a viscosity solution; the mean error against a brute-force grid envelope is ≈ 1.0 . The Hopf formula consistent with our verified `QuadraticL1` target carries a minus, $S(\mathbf{y}, t) = \sup_{\mathbf{p}} \{\langle \mathbf{p}, \mathbf{y} \rangle - J^*(\mathbf{p}) - \frac{t}{2} \|\mathbf{p}\|_2^2\}$, and since $J^*(P^\top \lambda) = \lambda^\top \gamma$ it collapses to a concave QP over the simplex,

$$S(\mathbf{y}, 1) = \max_{\lambda \in \Delta_m} \left\{ \langle P^\top \lambda, \mathbf{y} \rangle - \lambda^\top \gamma - \frac{1}{2} \|P^\top \lambda\|_2^2 \right\},$$

implemented in `src/targets.py:MaxPlus` and validated against the grid envelope to 1.1×10^{-3} (grid resolution). Restricting λ to the *vertices* recovers exactly the max-plus approximant Γ_K of the manuscript’s §3, a strict lower bound on S . Neither experiment appears in `main.pdf`; they live in `bin/not_in_paper/`.

2.3 [fixed] Under-trained, not under-sampled

A sweep over N at a fixed 300 epochs suggests a $4.6\times$ gain from more data. It is an artifact: 300 epochs on $4N$ points is also $4\times$ the optimizer steps. At a *fixed* step budget, $4\times$ the data buys only 6.7% ($n = 2$) and 9.9% ($n = 4$), whereas $4\times$ the steps on the *same* data recovers nearly the whole gain ($1.10 \times 10^{-3} \rightarrow 2.82 \times 10^{-4}$ at $n = 2$). Choosing S from a convergence probe under the real schedule: ψ ’s validation MSE at $(50, 100, 150) \times 10^3$ steps is $(2.62, 1.23, 0.83) \times 10^{-4}$ at $n = 2$ and $(1.54 \times 10^{-2}, 3.37 \times 10^{-3}, 1.83 \times 10^{-3})$ at $n = 8$. There is no true knee; the returns bend while cost stays linear, hence $S = 150\,000$. Re-running $n \leq 8$ improved ψ by $3.6\times$ to $293\times$ and cut prior RMSE $2\text{--}23\times$ for both routes, while the Route-1/Route-2 win count moved only from 8–4 to 7–4 with one tie: the ordering is carried by the protocol, not by the budget.

2.4 [fixed] Do not copy the paper’s decay cadence

The manuscript trains *full-batch*, so one of its “epochs” is one gradient step: 5×10^5 steps with the learning rate decayed $\times 0.1$ every 10^5 , i.e. every 20% of the budget, ending at 10^{-7} . That cadence presumes *exact* gradients. Ours are mini-batch estimates and need longer at the high learning rate. What tracks ψ ’s validation MSE is the number of steps spent at $\text{lr} = 10^{-3}$, not the total budget (`quadratic_l1`, $n = 16$):

S	decay fractions	steps at 10^{-3}	ψ val MSE
150 000	(0.5, 0.75)	75 000	6.99×10^{-2}
250 000	(0.3, 0.5, 0.7, 0.85)	75 000	6.30×10^{-2}
300 000	(0.5, 0.75)	150 000	7.83×10^{-3}

Doubling the high-learning-rate phase is worth $8\times$; adding 100 000 total steps while halving that phase is worth 11%. We therefore keep (0.5, 0.75) and reject the uniform 20% cadence. Early

stopping, added when the validation set was introduced, is also removed: we under-fit rather than overfit, and it truncated the G network on mini-batch noise (at $n = 16$ it stopped G at step 39 500 of 300 000).

2.5 **[fixed]** The width was ours, not the paper’s

All 24 notebooks of the four reported families use `hidden=256`, `layers=2`, `beta=5` at every dimension from 2 to 64, and all four table captions in the manuscript say “2 layers and 256 neurons”. The tiered width 64/128/256 was introduced by our own convention on 2026-07-08, motivated by a *performance* observation (the net is oversized at low n), never by an experiment. It is reverted. Beyond matching the manuscript, a fixed width removes a confound: with $N = 10\,000\,n$ and $|J| \sim 2n$ already growing with the dimension, a capacity that also jumps at $n = 8$ and $n = 32$ would leave an error-versus- n trend attributable to nothing. Cost is modest, since the CPU is overhead-bound: width 256 has $14\times$ the parameters of width 64 but takes only $2.7\times$ the time.

2.6 The initialisation is correct as it stands

Weight clipping pins $\approx 50\%$ of the inner weights to exactly 0 at the first step, because the default initialisation draws them negative. This costs nothing. At $n = 2$, ψ ’s validation MSE is 1.10×10^{-3} with the default, 6.25×10^{-3} with `|default|` (same magnitudes, no negatives, so the sign alone is responsible), and 9.51×10^{-2} with the log-normal initialiser that the upstream `init_weights` implements. Sign-clipping acts as useful sparsification. `init_weights` is deleted; it was dead upstream too.

2.7 Performance of the shared stack

`conjugate_samples` evaluated ψ twice per call (`model.scalar(x)` and then `model(x)`, which recomputes `scalar` internally); now one forward and one backward, $1.62\times$ faster with bitwise-identical output. `network.py:forward` built an unused second-order graph and set `requires_grad` on the *caller’s* tensor in place; both fixed, output identical, but the time saving is $1.01\times$, not a speedup worth claiming. Mini-batching (2026-07-08) remains the one large win, measured at $\approx 50\times$ over the original full-batch loop.

3 Cross-sections through the origin mislead in high dimension

The observation. At $n = 16$ (`quadratic_11`, $S = 250\,000$, $N = 240\,000$), the cross-section along the first axis shows ψ_θ sitting ≈ 1.45 *below* the exact ψ , and both recovered priors sitting 1.5–2.8 *above* the exact J near the origin — while the prior-recovery RMSE on random query points is 0.48 (Route 1) and 0.51 (Route 2), i.e. 1.5% and 1.6% relative. The pictures look bad and the numbers are good. Both are correct.

The mechanism. A cross-section along axis 1 fixes the other $n - 1$ coordinates at *exactly zero*. Under the uniform measure on $[-A, A]^n$, the set where $n - 1$ coordinates all lie within $\pm\varepsilon$ of zero has volume fraction

$$\left(\frac{\varepsilon}{A}\right)^{n-1},$$

which for $A = 5$, $\varepsilon = 0.5$ is 10^{-15} at $n = 16$ and 10^{-63} at $n = 64$. With $N = 240\,000$ training points, *not one* lies near that line. The cross-section is pure extrapolation; the RMSE is measured

on typical points, where the network fits. Low error and an ugly cross-section are not in tension: they are the same fact, read in two places.

This answers an open question in the manuscript. Section 4 currently says: “The errors for 64 dimensions in all the examples are consistently lower than expected. In contrast, it does not yield a good visual approximation of the cross sections. We are unable to explain the reason for this behavior in 64D cases yet; however, we think it is probably due to the way we sample the hypercube.” The guess is right, and the mechanism above makes it quantitative. The same remark disposes of the parallel puzzle about the $d = 64$ MSE “reducing significantly”: the reported error lives on the bulk of the cube, which is where the samples are.

Consequence. Cross-sections through the origin are a legitimate diagnostic at $n = 2$ and a misleading one at $n \gtrsim 8$. Figure 1 makes the point with a single network: the two panels are drawn from the *same* checkpoint at $n = 16$.

4 The diagnostic suite

`src/diagnostics.py` provides four in-distribution figures; `bin/plot.py` regenerates all of them from a saved checkpoint, for any family and dimension, without retraining (`bin/plot.py --family quadratic_l1 --dim 16 --open`).

1. **Conditional cross-section** (`plot_conditional_cross_section`). Vary x_1 ; draw the remaining coordinates from the training distribution; repeat for six draws; centre every curve by subtracting its own value at $x_1 = 0$, so the six additive constants contributed by the frozen coordinates do not separate the curves. What remains is the x_1 -dependence on the data manifold. For a *separable* prior the centred exact curves collapse onto one line, so any spread among the learned curves is the network’s error; for a non-separable prior they do not collapse and the figure must be read per draw. Both routes are shown, with Route 1’s six inversions batched into a single solve.
2. **Typical ray** (`plot_typical_ray`). Profile along $t \mapsto t \mathbf{1}/\sqrt{n}$, clipped to the box. Most of such a ray lies in the bulk; only the neighbourhood of the origin is atypical.
3. **Predicted versus true** (`plot_pred_vs_true`). $\hat{J}(\mathbf{x})$ against $J(\mathbf{x})$ over the test set, with the identity line, one panel per route. Dimension-agnostic; shows bias and spread together; cannot be gamed by the choice of a slice.
4. **Two conjugate scatters**, one per network. The *prox scatter* (`plot_prox_scatter`) plots the learned $\nabla\psi_\theta$ against the exact prox_J , coordinatewise, on the training box. It is a diagnostic of the *shared* first network: Route 1 inverts $\nabla\psi_\theta$, and Route 2 is fitted to its image, since G is trained on $\mathbf{y}_k = \nabla\psi_\theta(\mathbf{x}_k)$ with targets $G_k = \langle \mathbf{y}_k, \mathbf{x}_k \rangle - \psi_\theta(\mathbf{x}_k)$. It therefore bounds *both* routes. The *preimage scatter* (`plot_preimage_scatter`) is Route 2’s own: it plots $\nabla G(\mathbf{x})$ against the analytic preimage $\mathbf{y}^*(\mathbf{x})$, testing the identity $\nabla G = (\nabla\psi)^{-1}$ on which Route 2 rests. It costs one forward pass.

An error profile binned by $\|\mathbf{x}\|_\infty$ was built and *removed*: in high dimension the maximum of n uniform coordinates concentrates near a , so the bin variable has almost no dynamic range (2.6 to 3.95 at $n = 16$) and the figure reads as noise.

quadratic_l1_16D (axis cross-section: IN-DISTRIBUTION ONLY AT d=2)

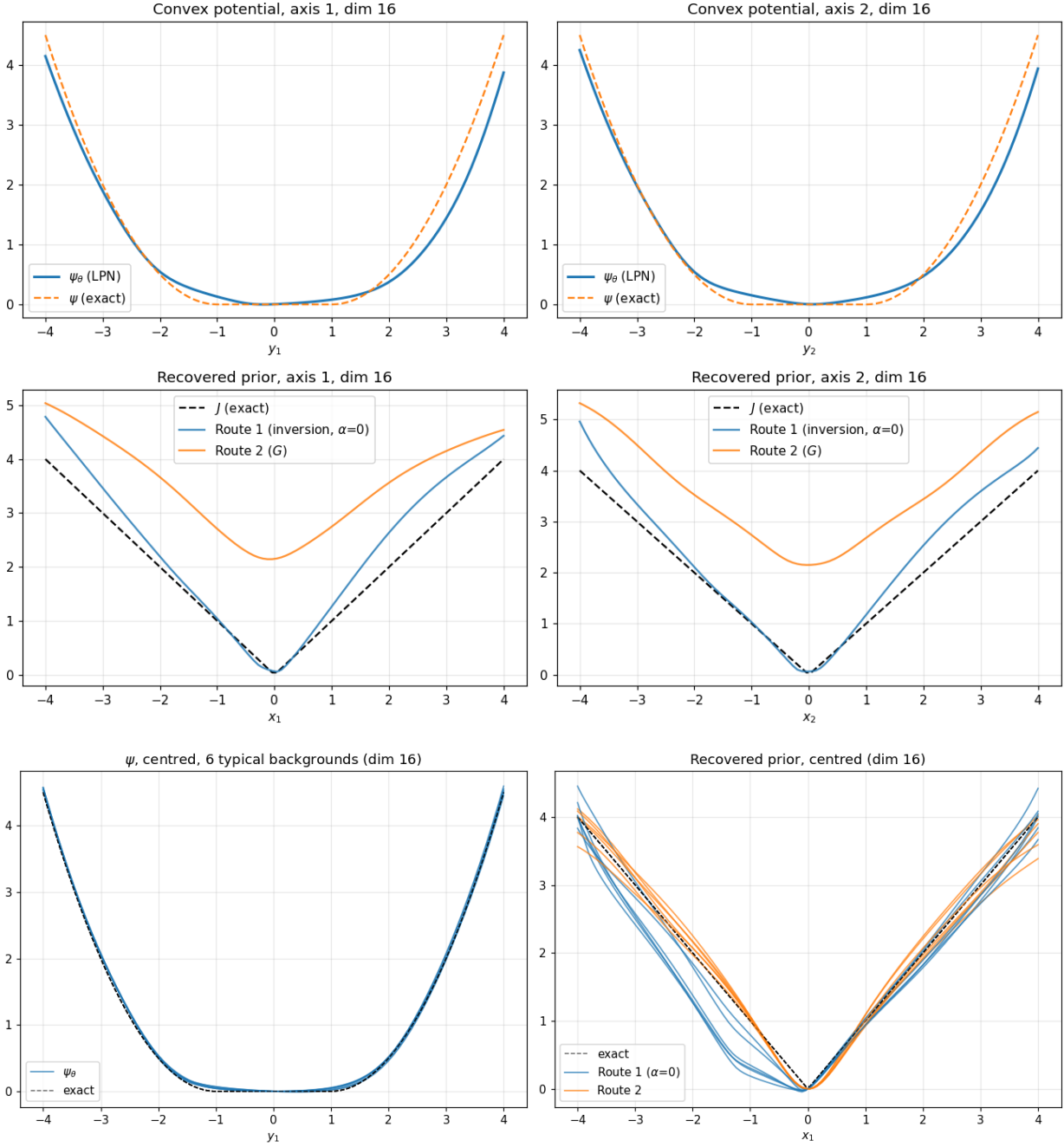


Figure 1: Same weights, opposite verdicts (quadratic_l1, $n = 16$, $S = 250\,000$). **Top:** the usual cross-section, holding the other 15 coordinates at exactly zero. ψ_θ appears offset by ≈ 1.45 and both recovered priors are wrong by up to 2.8. **Bottom:** the conditional cross-section, which differs only in drawing those 15 coordinates from the training distribution and centring each curve at $x_1 = 0$. ψ_θ lies on the exact curve and both routes track J to within ≈ 0.3 . The reported RMSE is 0.196 (Route 1) and 0.284 (Route 2), i.e. 0.61% and 0.89% relative. The network was never bad; the top plot plots it where no data exists.

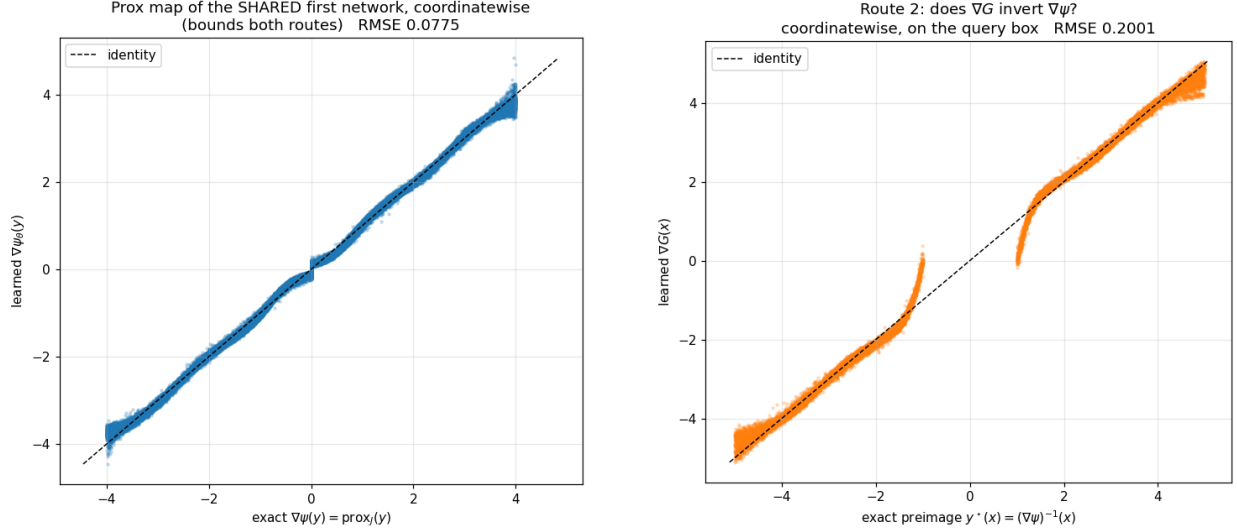


Figure 2: The two conjugate scatters at $n = 16$. **Left:** the shared first network. $\nabla\psi_\theta$ lies on the identity against $\text{prox}_J = \text{soft}(\cdot, 1)$, reproducing even the dead zone at the origin; RMSE 0.078. **Right:** Route 2’s defining identity, $\nabla G(x)$ against $y^*(x) = x + \text{sign}(x)$. It holds wherever it is testable. The gap for $|y^*| < 1$ is structural, not an error: the preimage map never takes values there, so ∇G has nothing to fit — the same hole that afflicts **NegL1**, seen from the conjugate side.

5 Retractions

Recorded because the numbers moved substantially, and a reader who sees that should know which claim died and what killed it.

[retracted] “The preimage lies far outside the training box, $\|y\|_\infty \approx 19$.” The exact preimage obeys $\|y^*\|_\infty \leq 5$ at every n . The 19 was a diverging iterate whose size is set by $\text{lr} \times \text{iters}$.

[retracted] “A zero prox residual certifies that a preimage exists.” It does not; see §1.

[retracted] “Route 1 is more honest, because it detects its own failure.” It is either divergent ($\alpha = 0$ with an under-trained ψ_θ) or silently biased ($\alpha > 0$).

[retracted] “Route 1 at $\alpha = 0$ is intrinsically ill-posed.” The divergence was a symptom of an under-trained ψ_θ . The objective is coercive exactly when $x \in \text{range}(\nabla\psi_\theta)$, and an under-trained $\nabla\psi_\theta$ saturates below the corner values of the query box. With $S = 150\,000$ no configuration diverges; **quadratic_11_8D** selects $\alpha = 0$ with $\|y\|_\infty = 5.20$ against the exact bound 5. Keep the two- α rule as cheap insurance, but do not claim the unregularized inversion is broken.

[retracted] “The **NegL1** origin hole is a Route-2 coverage defect.” Route 1, which never touches G , is hit as hard (mean error $0.029 \rightarrow 0.342$ versus Route 2’s $0.025 \rightarrow 0.276$ at $n = 2$). The shared cause is that a smooth Softplus cannot represent the kink of ψ at $y_i = 0$, where $\partial\psi(0) = [-1, 1]$.

[retracted] “**exp_4_1_3**’s target is correct” (Phase 1); see §2.2.

A rediscovery, not a discovery. The α regularizer and the existence problem it addresses were already recorded in the 2026-07-08 audit, which noted that an LPN “may not be invertible or even surjective” and that Fang et al. add $\frac{\vartheta}{2} \|\mathbf{y}\|_2^2$ to force strong convexity. The 2026-07-09 work re-derived this from the code. The lesson is procedural: read the audit before re-auditing.

6 What we can claim about Route 1 versus Route 2

On the symmetric protocol of §1 the two routes are comparable at every dimension we have run, with no gap exceeding $2.6\times$. Route 1 wins where the prior is smooth or separable, Route 2 wins narrowly on `minplus`. Nothing supports “Route 2 dominates”; every earlier table that suggested so had Route 1 handicapped, first by the training box and then by a mismatched evaluation set.

The defensible claim is about *cost and well-posedness*, not accuracy: Route 2 attains comparable error with one forward pass, no regularizer, no per-query optimization, and no well-posedness question, whereas Route 1 needs an optimization per query and a parameter α that trades divergence against an $O(\alpha)$ bias. Note also that the residual magnitudes are *not* comparable across routes: Route 2’s is uniformly an order of magnitude larger while its RMSE is competitive, because the residual measures preimage consistency, not prior accuracy.

The live numbers are in `numerics/logs/summary_final.csv`. Every table predating the current architecture is superseded; `changes.txt` lists which and why.

7 Open items

- [open] **Phase 5**, the error-versus- K decay study for Γ_K — the experiment referee R2.3 actually asked for. Still unbuilt. `MaxPlus.maxplus_approx()` now supplies Γ_K against an exact reference, so the ground truth is in hand.
- [open] **Is width 256 right?** Untested, and now testable: every previous measurement was taken at a budget too short to converge, so a width ablation would have measured which width *trains fastest*, not which *fits best*. The revert is justified by consistency with the manuscript, not by evidence that 256 is optimal.
- [open] **High dimensions.** $n \geq 16$ has not been run under the current protocol. Cost is modest (≈ 10.4 and 17.7 minutes per configuration at widths 128 and 256), but N/P and $|J|$ both move with n , so the trend needs care.
- [open] **The max-plus ports** use the D2 MSE pipeline, whereas the notebooks trained with LPN’s proximal-matching loss on a decreasing γ schedule. They therefore measure something different. Deliberate; revisit if the training scheme becomes the object of study.
- [open] **$N = 10P$ instead of $N = 10\,000\,n$.** Under a step budget the cost is `steps` $\times$ `batch`, independent of N , so a constant $N = 10P \approx 1.4\text{M}$ would make width, depth, N and S all constant and leave the dimension as the sole variable. Cleaner, not adopted: one change, not two.